◈ PART A — THEORY

1. What is Statistical Distribution?

A statistical distribution describes how values of a variable are spread or distributed. It shows the frequency or probability of different outcomes. Examples include Normal, Binomial, Poisson, and Log-Normal distributions.

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2. What is a Q-Q Plot?

A Q-Q (Quantile-Quantile) Plot compares your dataset with a theoretical distribution (usually Normal).

- If points lie on a straight line → data is normally distributed
- If curved → data is skewed or non-normal

Used to check **normality assumption**.

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3. Discrete vs Continuous Distributions

Feature	Discrete	Continuous
Values	Countable (0,1,2)	Infinite (1.5, 2.7)
Example	Number of purchases	Transaction amount
Distribution	Binomial, Poisson	Normal, Log-normal

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4. Bernoulli Distribution

Represents a single trial with two outcomes:

- Success (1)
- Failure (0)

Example: Transaction success/failure

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5. Binomial Distribution

Represents number of successes in **n trials**.

Example: Number of successful transactions in a week.

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6. Log-Normal Distribution

If the logarithm of a variable is normally distributed, then it follows Log-Normal distribution.

Used for:

- Income
- Transaction amounts (always positive & skewed)

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7. Power Law Distribution

Heavy-tailed distribution where:

- Few customers spend a lot
- Most spend small amounts

Common in e-commerce behaviour.

8. Box-Cox Transform

Used to:

- Normalize skewed data
- Stabilize variance

Formula transforms data into a more normal shape.

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9. Poisson Distribution

Used for counting events over time.

Example:
Number of transactions per day.

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10. Z-score Probability

Measures how far a value is from the mean:

$$Z = \frac{X - \mu}{\sigma}$$

Used to calculate probabilities using standard normal distribution.

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11. PDF vs CDF

PDF	CDF
Probability Density	Cumulative Probability
Shows likelihood	Shows total probability up to X
Curve	Increasing curve

What is Bernoulli Distribution?

A **Bernoulli Distribution** is used when:

- There is **only ONE experiment**
- It has **ONLY TWO outcomes**

Why mean = probability?
Because in Bernoulli:

$$p = \frac{\text{Number of Successes}}{\text{Total Trials}}$$

And mean of 0s and 1s is exactly that!

Mean: (1+0+1+1+0)/5=0.6

Binomial Distribution (Weekly Transactions)

Out of n trials, how many successes can occur?

📌 What is pmf?

PMF = Probability Mass Function

👉 It gives probability of exact number of successes

📌 range(0,8) means:

We calculate probability for:

Successes (k)	Meaning
0	No success in week
1	1 success
2	2 successes
...	...
7	All 7 days successful

📌 **What this calculates:**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$